
ABC
SMITHY
KEEP RIGHT
LABORATORY
CBC
BMP
COAG
Ddx
ACLS
V-tach
FILL
Hx
NITRO
WAIT TIME:
A COUPLE HOURS
Stays fresh
WEEKS
Fried chicken
now More Biscuits
Good the NEXT DAY
Try re-heated!
CAUTION
CAUTION
GRACE orphanage
Armor Plate Co.
1st St
1st St
2nd St
2
17
3
Armor Plate Co.
MoNa B's
Nitro shop
Knock 3 times
back in 5
Do not over-FILL
take a prize
CLOP
CLOP
CLOP
CAUTION
chicken

1.3 – Acute Coronary Syndromes SOA

1) Broken heart string:
acute coronary syndrome (ACS - **UA, NSTEMI and STEMI**)

7(S) Angina anvil:
patients with ACS may present with **ischemic chest pain** (angina) (CC)

8(O) “ABC” Smithy:
assess **airway, breathing, and circulation** first with suspected ACS (e.g. pulse, blood pressure, prepare for bipap or intubation) (**Vitals**)

9(P) The king’s ivy:
establish **IV access** in patients with suspected ACS (**1**)

10(P) The king’s tubing:
give O2 with a goal of maintaining SpO2 >90% (**2**)

11(O) The king’s glowing red scepter:
monitor SpO2 (goal >90%) (**Vitals**)

12(P) The king’s monitoring guard:
place patients on **cardiac monitor** and **continuous pulse oximeter (3)**

13(P) ACLS snack cart and V-tach dog:
new arrhythmias (Afib, VT, AV block) should raise suspicion for ACS → **use ACLS protocol to treat sustained ventricular arrhythmias (4)**

20(P) ECG flags with quarter clock:
initial ECG is **often nondiagnostic** → **repeat ECG every 10-15 minutes** if suspicion for ACS remains high (**5**)

21(P) ASA umpire feeding horse:
give **325 mg ASA (chewed)** immediately in ACS (**6**)

22(S) Historian’s anvil:
focused HPI → confirm presenting symptoms, characterize pain, assess associated symptoms (**HPI**)
23(S) Historian’s atherosclerotic hat:
focused PMHx + SHx → ask about risk factors for **CAD** (HTN, HLD, DM, smoking, + family hx, prior MI); ask about **drug use** (e.g. cocaine) **(PMHx+SHx)**

24(S) Historian spilling red ink:
ask about **contraindications to thrombolytic therapy** (e.g. previous ICH, cerebral AVM, intracranial neoplasm, ischemic stroke/head trauma within 3 months, suspected aortic dissection, active bleeding or bleeding diathesis) **(PMHx+HPI)**

25(S) Historian's medication bag:
review medications, especially **phosphodiesterase-5 inhibitor use** (e.g. sildenafil) (contraindication for nitrate therapy) **(Meds)**

26(O) Physical guard:
focused physical exam → cardiopulmonary exam, signs of hypoperfusion/shock, signs of HF **(PE)**

27(O) Whistling with S3 slushie on S4 chair:
S3, S4, or new mitral regurgitation murmur may be signs of myocardial ischemia/infarction **(PE)**

28(O) Fainting next to shock sword:
assess for **signs of cardiogenic shock** (e.g. hypotensive, Ill-appearing, cool, pale, diaphoretic, lethargic, disoriented) → **aggressive treatment! (Vitals+PE)**

29(O) Failing heart balloon:
assess for **signs of heart failure** (e.g. hypotension, elevated JVP, bibasilar crackles, S3) → aggressive treatment! **(Vitals+PE)**

34(O) Brain diagram:
focused neuro exam (assess baseline neurologic function and assess for deficits that might **preclude the use of thrombolytic therapy) (PE)**

35(O) Normal person:
physical exam may be normal with ACS **(PE)**

36(O) Medieval lab:
order **labs** (e.g. CBC, BMP, coags, cardiac biomarkers) **(Labs)**

44) DDx clipboard:
For further workup of Acute Coronary Syndrome, see Chest Pain DDx

45) "MONA-B'S":
initial management of ACS includes **M**orphine, **O**xygen, **N**itrates, **A**spirin, **B**eta-blocker and **S**tatin

46(P) "Nitro Shop":
Give a **short-acting nitrate** (sublingual nitrogen) for ongoing chest discomfort, hypertension or heart failure (if no contraindications!) (**8**)

47(P) “Knock 3 times/Back in 5”:
give 3 sublingual nitroglycerine tablets, spaced 5 min apart as needed (if no contraindications!) (**9**)

48(P) Ivy next to sad minstrel:
add **IV nitroglycerine for persistent symptoms** (if no contraindications!) (**10**)

49) Dilated venous pants and arterial sleeves:
nitroglycerine causes **venous dilation** and (less) arterial dilation → ↓preload and ↓afterload to ↓myocardial O2 demand

50) Dilated crown:
nitroglycerine ↑**coronary vasodilation** to ↑myocardial O2 supply

51) “do NOT overFILL” with nitro:
nitrate therapy is **contraindicated with PDE-5 inhibitor** use (e.g. sildenaFIL) within 24 hrs

52) “Caution fall” with nitro:
nitrate therapy is **contraindicated in hypotensive patients**

54(P) Muted beta bugle:
give **beta-blocker** (e.g. metoprolol 25mg PO) **within the first 24 hours** of diagnosis of ACS (if no contraindications!) → reduce infarct size and early mortality (**11**)

55) “Caution fall” behind failing bradycardic bugler:
beta-blocker therapy is **contraindicated in patients with bradycardia, signs of heart failure, or signs of hemodynamic compromise**

56) Choking with bugle:
beta-blocker therapy is **contraindicated in patients with severe reactive airway disease** (nondihydropyridine calcium channel blockers like diltiazem can be used instead)

57(P) **M** in “MONA B’s”:
give **morphine** if chest pain is not adequately controlled with nitrates (if not contraindicated!) (**12**)

59(P) Intense “**S**” in “MONA B’s”:
give a **high-intensity statin** at diagnosis (atorvastatin 80 mg) (**13**)

60(P) Fainting with “PRESSOR” tank:
give **pressors** for patients with **cardiogenic shock** (e.g. norepinephrine) (**14**)

2) Unstable anvil:
Unstable angina (cardiac ischemia without infarction) presents as chest pain that increases in frequency, duration, or intensity from baseline; chest pain occurring at rest; or new-onset severe chest pain

3) Unstable anvil with fallen ST sign:
unstable angina can occur **with or without ECG changes indicative of ischemia** (e.g. ST segment depression, transient ST elevations, or new T wave inversions)

19) Fallen ST sign next to NSTEMI minstrel and UA anvil:
UA and **NSTEMI** may show **ischemic changes on ECG** (e.g. ST depression, transient ST elevation, new T wave inversions - often transient in UA and PERSISTENT in NSTEMI) (ECG may be normal)

76(P) "**A**rmor **Plate**" sign with falling ST:
In **UA/NSTEMI** start **dual anticoagulation therapy** with aspirin and a P2Y12 (ADP) receptor blocker (ticagrelor, prasugrel, or clopidogrel) (**23**)

77(P) "**A**rmor **Co**" sign with falling ST:
for **UA/NSTEMI** patients, start **anticoagulation** (usually unfractionated heparin, consider enoxaparin or fondaparinux if no invasive procedure planned) (**24**)

78) Timmy at GRACE orphanage:
TIMI and **GRACE** risk scores are used in UA/NSTEMI to **determine risk of complications** (e.g. recurrent MI, recurrent angina requiring revascularization, and death) → pursue invasive vs conservative approach

79(P) Balloon for **unstable** orphan in pain:
Indications for **immediate angiography in UA/NSTEMI** include **hemodynamic instability**, heart failure, ongoing/recurrent rest angina, acute MR/VSD, sustained VT, or dynamic ST/T wave changes (**25**)

80(P) Elevated balloon next to orphanage:
UA/NSTEMI patients at **intermediate to high risk** (abnormal vital signs, heart failure, older age, increased cardiac biomarkers, EKG abnormalities, hx of CAD, risk factors for CAD) undergo **angiography within 24-48 hrs** (**26**)

81(P) Low treadmill next to orphanage:
low risk UA/NSTEMI undergo **stress testing** prior to discharge (**27**)

31) **Inferiorly dirty** dress:
inferior MI (occlusion of the right coronary distribution → ST elevations in leads II, III and aVF)

32(O) **Fainting** with **blue braids** and **clear lungs**:
assess for signs of **RV failure** (especially with inferior MI) (e.g. triad of **hypotension, clear lungs, elevated JVP**) → treatment to maintain RV preload! **(Vitals+PE)**

33(O) Dangling heart watch:
assess for **bradycardia** (especially with **inferior MI** due to increased vagal tone) **(Vitals)**

37) T-bones in "a couple hours":
cardiac **troponins** (sensitive and specific for MI) are detectable **2-3 hours** after symptom onset

38) T-bones are "fresh for weeks":
cardiac **troponins** peak at ~12 hours and can stay **elevated for 10-14 days**

39) "chiCKen with More Biscuits":
CK-MB may also be elevated in MI (less sensitive and specific)

40) "Good the next day":
CK-MB begin to rise 4-6 hours after infarction and **return to baseline within 36-48 hours**

41) "Try re-heated":
theoretically, **CK-MB** levels can be useful for **detection of reinfarction** (levels return faster to baseline) but **trops are still preferred** (more sensitive and specific)

42(P) Repeating T-bone and ECG flags:
repeat troponin levels and ECG every 6 hours (monitor for signs of worsening ischemia or improvement) (**7**)

53) "No right turn" with nitro:
nitrate therapy should be **avoided in patients with an right ventricular MI** (↓ RV filling)

58) "Caution fall" + "No right turn" with "M":
morphine is **contraindicated in cardiogenic shock** (↓ BP) **and right-sided MI** (↓ RV filling)

62) "Caution fall" + "No right turn" with nitro cart and loop horn:
nitro and diuretics are **contraindicated in cardiogenic shock** (↓ BP) **and right ventricular MI** (↓ RV filling)

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64(P) Atropine alice with low heart watch:
patients with **symptomatic bradycardia** following acute MI can be treated with **atropine** (**17**)

65(P) “90” balloons:
perform **coronary angiography within 90 minutes** of presentation (**18**)

66) Dyed coronary hat:
coronary angiography is performed during cath procedure (identify location of stenosis)

67) Dry and wet stent arms:
PCI involves placement of **bare metal stents** (BMS) or **drug-eluting stents** (DES)

68(P) Cabbage with clod on left root and multiple roots:
Indications for **CABG** include **L main coronary artery stenosis and severe multivessel disease** (**19**)

69) Popping out of thrombotic hay:
Thrombolytic (aka fibrinolytic) **therapy** (e.g. with tenecteplase) is an alternative to PCI to restore coronary blood flow (PCI is preferred)

70(P) **2 fingers** popping out of hay:
If possible, **transfer to PCI-capable hospital within 120 minutes** of first medical contact (otherwise may proceed with thrombolytic therapy if no contraindications) (**20**)

71) Historian spilling red ink:
ask about **contraindications to thrombolytic therapy** (e.g. previous ICH, cerebral AVM, intracranial neoplasm, ischemic stroke/head trauma within 3 months, suspected aortic dissection, active bleeding or bleeding diathesis)

4) Broken heart string with fallen ST sign:
non-ST elevation MI (NSTEMI) presents with new onset ischemic chest pain **with or without ECG changes indicative of ischemia** (e.g. ST segment depression, transient ST elevations, or new T wave inversions)

5) T-bones next to broken heart strings:
NSTEMI and **STEMI** can present with **elevated cardiac biomarkers** (e.g. troponins)

14(O) ECG reins:
obtain a **12 lead ECG** within 10 minutes of arrival (**ECG**)

15(O) Ischemic skull on ECG flag:
look for ischemic changes on ECG (e.g. ST depressions, ST elevations, T wave inversions, peaked T waves, Q waves, new LBBB) (**ECG**)

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6) Broken heart string with elevated ST sign:
ST elevation MI (STEMI) is characterized by ischemic chest pain, **persistent ST elevations** and cardiac biomarkers

30) Fainting fan with blue braids, clear lungs, and inferior dress:
signs of **right ventricular MI** include **inferior STEMI** (II, III, aVF), **hypotension, clear lungs, elevated JVP**

16) 2 contiguous 1st ST signs:
A **STEMI** is defined as **≥1mm ST elevation in 2+ contiguous leads** or ≥2mm ST elevation in V2 and V3

17) 2nd ST with V2 and V3 flags:
A **STEMI** is defined as **≥1mm ST elevation in 2+ contiguous leads** or ≥2mm ST elevation in V2 and V3

18) Bundle of sticks next to STEMI minstrel:
EKG with **new left bundle branch block** may suggest myocardial ischemia/infarction (**treat patient like they have a STEMI**)

72(P) "**A**rmor **Plate**" sign with elevated ST:
for **STEMI**, start **dual antiplatelet therapy (DAPT)** with aspirin and a P2Y12 (ADP) receptor blocker (clopidogrel, ticagrelor, or prasugrel) (**21**)

73) "Take a prize" with PCI balloons:
ticagrelor and **pras**ugrel are preferred **antiplatelet agents** for STEMI patients undergoing **PCI**

74) "Clop" with kid popping out of hay:
clopidogrel is the preferred **antiplatelet agent** for STEMI patients undergoing **thrombolytic therapy**

75(P) "**A**rmor **Co**" sign with elevated ST:
for **STEMI**, start **anticoagulation** (heparin, bivalirudin, fondaparinux, or enoxaparin) (heparin preferred for PCI) (**22**)

61(P) Wet nitro cart and loop horn:
patients with signs of **left heart failure** (e.g. pulmonary edema) should be treated immediately with **nitrates**(e.g. nitroglycerin for ↓ preload + ↓ afterload) and an **IV loop diuretic** (e.g. furosemide for diuresis and venodilation) (**15**)